Yifan Zhao

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Education

Massachusetts Institute of Technology — Cambridge, United Sates
PhD in Medical Engineering and Medical Physics — *2020 – present*
- Thesis Advisor: Dr. Peter J. Park

McGill University — Montreal, Canada
Honours BSc in Computer Science and Biology — *2017 – 2020*
- First Class Honours with Distinction
- Thesis Advisor: Dr. Abigail R. Gerhold

Research & Teaching Experience

PhD Student, Park Lab
Cambridge, United States (remote) — *Dec '20 – present*
- Copy number analysis on single-cell DNA-sequencing samples of human brain

Summer Research Intern, Li Lab
Montreal, Canada — *May '20 – Aug '20*
- Developed single-cell Embedded Topic Model (scETM), a Bayesian inference model for single-cell RNA-seq data integration and transfer learning with interpretable latent dimensions
- Benchmarked 7 state-of-the-art single-cell RNA-seq clustering methods

Honours Research Student, Gerhold Lab
Montreal, Canada — *May '19 – May '20*
- Developed CentTracker, an automated analysis pipeline for centrosome tracking and pairing
- Conducted *in situ* live-cell imaging experiments of *C. elegans* germline stem cells

Research Assistant, Yamanaka Lab
Montreal, Canada — *Dec '18 – Apr '19*
- Automated dynamic MiSeq sequencing data analysis
- Quantified and classified Cas9 RNA-guided endonucleases off-target sites to identify the clonal selection patterns during cancer progression in mice models of ovarian cancer

Undergraduate Teaching Assistant, McGill University
Montreal, Canada — *Jan '19 – Dec '19*
- MATH 240 Discrete Structures (Fall 2019)
- MATH 324 Statistics (Winter 2019)

Publications & Talks

1. **Zhao Y.**†, Cai H.†, Zhang Z., Tang J.*, Li Y.* (2021). Learning interpretable cellular and gene signature embeddings from single-cell transcriptomic data, bioRxiv 2021.01.13.426593 (in revision; †equal contribution, *co-corresponding authors).

2. Zellag M. R., **Zhao Y.**, Poupart V., Singh R., Labbé J-C., Gerhold A. R. (2021). CentTracker: a trainable, machine learning-based tool for large-scale analyses of *C. elegans* germline stem cell mitosis, Molecular Biology of the Cell, Jan 27:mbcE20110716.

Awards & Scholarships

- Jacqueline Johnson Desoer Science Undergraduate Research Award (2020)
- Sheila Ann MacInnis Grant Undergraduate Research Award (2019)
- E Gordon Edwards Biology Award (2019)
- Faculty of Science Scholarship (2019)
- Dean's Honour List (2018)
- James McGill Scholarship (2017-2020)
- Governor General's Academic Medal (2017)
- National Biology Scholar with Distinction (2017)
- British Columbia International Student Ambassador Scholarship (2017)

Proficiencies

- **Programming Languages**: Python, C++, R
- **Natural Languages:** Chinese, English, French, Wenzhounese
- **Laboratory techniques**: Confocal spinning disk microscopy, *C. elegans* care and husbandry